GGG-D-694a
MAY 27, 1960
SUPERSEDING
GGG-D-00694(NAVY-SHIPS)
December 10, 1957

FEDERAL SPECIFICATION

DRILL, MASONRY, ROTARY DRIVEN

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of masonry drill, carbide tipped, core type, for use in drilling holes in plain or reinforced concrete and other type of masonry used with portable power rotary drills.

1.1.1 *Federal specification coverage.*—This specification does not include all types, classes and sizes of masonry drills covered by the title of this specification, but includes only those most commonly used by the Federal Government.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-P-15424—Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specification and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

American Society for Testing Materials Standard:

ASTM Designation C 131–47 — Method for Test of Abrasion of Coarse Aggregate by use of the Los Angeles Machine.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

3. REQUIREMENTS

3.1 Material.—The body of the drills shall be steel bar stock, or steel tubing suitable for its intended use. The shank of the drill shall be made from steel bar stock, the body of the drill shall have carbide tipped inserts for cutting (drilling) operation.

3.2 Identification markings. — Each drill shall be stamped or otherwise permanently marked with its nominal size and the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.3 Design.—The drills shall have a right-hand lead with spiral grooves (worm type), cut on the periphery of the body or may be furnished without spiral grooves with openings on the body, designed for the removal of material during drilling operation. The body shall be bored approximately 80 percent of its length and shall be provided with a slot or other openings for removal of core. The body shall be so designed with slots to receive the carbide-tip inserts which shall be properly brazed to permit efficient drilling of the material. When shanks are furnished separate from the body, threads shall be provided in body to allow for assembly of the shank with the body. On drills ¾ inch in diameter and larger, there shall be a collar provided on the shank. The shank, when assembled to the body, shall be concentric within a maximum of 0.025 inch run out for true alignment. The design of the shank shall be straight and of a diameter and length to function efficiently in performing satisfactory work within its size range. Drills shall be furnished with a removable single carbide tipped pilot drill when specified (see 6.1). The drills shall be similar to figure 1 and shall conform to table I for the nominal size specified (see 6.1).

3.3.1 *Illustration.*—The illustration shown herein is for the convenience of identification and is not intended to preclude the purchase of drills which are otherwise in accordance with the requirements of this specification.

3.4 Hardness.—The carbide insert tips shall have a minimum hardness of 92 on the Rockwell "A" scale. The body of the drill shall be properly tempered and hardened to not less than 10 nor more than 50 Rockwell "C" scale (see 4.5.1).

3.5 Finish.—Surfaces, other than carbide cutting inserts, shall be machined finish in accordance with good commercial shop practice.

3.6 Workmanship.—Workmanship shall be first class in every respect. Masonry drills and shanks shall be free from rust, laps, cracks, and other imperfections which may impair their serviceability or durability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Place of inspection.—Inspection shall be performed at the place of manufacture.

4.3 Sampling for acceptance inspection.

4.3.1 *Lot.*—All masonry drills of the same size, offered for delivery at one time, shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

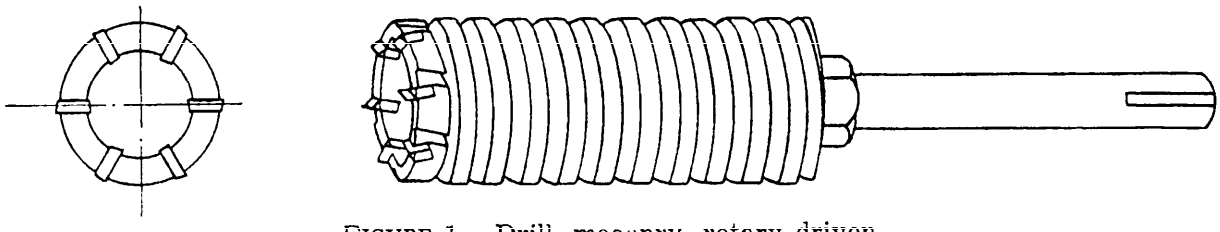


FIGURE 1.—Drill, masonry, rotary driven.

TABLE I.—*Drills, masonry, carbide tipped, core type, rotary*

Size	Body diameter	Carbide tip diameter [1]	Shank diameter	Overall length
Nominal	+0.000 −0.030		±0.010	Minimum
Inches	*Inches*	*Inches*	*Inch*	*Inches*
1/2	0.500	0.525	0.375	6
9/16	.562	.587	.375	6
5/8	.625	.650	.375	8
5/8	.625	.650	.500	6
11/16	.687	.712	.375	8
11/16	.687	.712	.500	6
3/4	.750	.775	.500	8
13/16	.750	.812	.500	8
7/8	.832	.900	.500	10
1	.958	1.025	.500	10
1 1/8	1.085	1.155	.500	10
1 1/4	1.206	1.280	.500	10
1 3/8	1.331	1.405	.500	10
1 1/2	1.455	1.530	.500	11
1 3/4	1.750	1.780	.500	11
1 7/8	1.750	1.905	.500	11
2	2.000	2.030	0.500 or .750	11
2 1/8	2.125	2.157	.750	12
2 1/4	2.250	2.281	.750	15
2 3/8	2.375	2.406	.750	15
2 1/2	2.500	2.531	.750	15
2 5/8	2.625	2.656	.750	15
2 3/4	2.750	2.781	.750	15
2 7/8	2.875	2.906	.750	15
3	3.000	3.312	.750 or 1.000	15
3 1/4	3.250	3.281	.750 or 1.000	15
3 1/2	3.500	3.531	.750 or 1.000	15
3 3/4	3.750	3.781	.750 or 1.000	15
4	4.000	4.031	.750 or 1.000	15

[1] Drills with a carbide tip diameter of up to 2 inches drill diameter shall have a tolerance of $^{+0.010}_{-0.000}$ drills with a carbide-tip diameter 2 inches and above drill diameter shall have a tolerance of $^{+0.031.}_{-0.000.}$

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:42Z
Check the source to verify that this is the currentversion before use.

considered a lot for purposes of sampling and inspection.

4.3.2 *Sampling for visual and dimensional examination.*—From each lot, a sample shall be selected at random in accordance with Military Standard MIL–STD–105, at inspection level II, for the examination of 4.4. Lots shall be accepted or rejected in accordance with Acceptable Quality Level (AQL) = 1.5 percent defective.

4.3.3 *Sampling for lot acceptance tests.*—A random sample of drill shall be selected from each lot in accordance with Military Standard MIL–STD–105 at inspection level L–7 for the tests of 4.5. The Acceptable Quality Level shall equal 2.5 percent defective. However all the sample sizes shall be used, and the acceptance number shall be zero until the AQL permits one or greater.

4.4 **Examination.** — Each of the sample drills selected in accordance with 4.3.2 shall be examined to verify conformance with all the requirements of this specification which do not involve tests. Any drill in the sample containing one or more defects shall be rejected and if the number of defective drills in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.5 **Test procedures.**

4.5.1 *Hardness test.*—Hardness tests shall be conducted in accordance with method 243 of Federal Test Method Standard No. 151 and the value thus obtained shall comply with the requirements specified in 3.4. Impressions shall be taken on the body of the drill and on the carbide tip inserts.

4.5.2 *Drilling test.*—Masonry drills shall be tested on an upright drilling machine with positive feed, and operated at the speeds and feeds specified in table II. The minimum depth that drill is to be tested shall be 4 inches for a minimum of three holes for each size drill as specified in table II. Drilling tests shall be performed dry and without use of an air jet. When tested as specified in table II, there shall be no indication of excessive wear on body and on cutting edges of carbide inserts, no indication of bending, deforming or fractures in the body of the drill, no signs of spalling, chipping, or cracking of cutting edges of the inserts.

TABLE II.—*Drilling test*

Size of drill	Revolutions per minute (r. p. m.)	Inch feed per revolution
Inches		
½ to ¾	300 to 400	0.008
13/16 to 1⅞	200 to 300	.008
2 to 4	100 to 200	.008

4.5.2.1 *Test block.*—The test block shall be of molded masonry having a minimum thickness of four inches (the depth of one test hole). The block shall be thoroughly cured a minimum of five days in a steam box under controlled temperature and pressure conditions or shall be adequately moist cured for a minimum period of 28 days after stripping the block from the molds. Keeping the block buried in moist sand at room temperature (70 to 75 degrees Fahrenheit) or keeping in a fog room at 95 percent humidity and at room temperature shall be considered adequate moist curing. The following material shall be required for the test block:

(*a*) Two parts sharp washed sand (medium coarse).

(*b*) One part cement.

(*c*) Sufficient water to make mixture of stiff consistency (approximately 5½ gallons to one 94-pound sack of cement).

(*d*) Two parts coarse, hard aggregate, ⅜ to ½ inch size. The aggregate shall be hard flint bluestone, "agate" pebbles, or other equivalent stone or rock which will lose not more than 18 percent weight if tested

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

in accordance with test C–131, test sample grading "A", of the "Standard Method of Test for Abrasion of Coarse Aggregate by use of the Los Angeles Machine" as specified in the current issue of the Book of American Society for Testing Materials (ASTM) Standards. (Crushed fine grained igneous rock or stone, either basalt or diabase, shall be considered as equivalent aggregate provided it is of a quality which will meet the abrasion test requirements.)

The drill supplier shall furnish the inspector with written certification that the coarse aggregate used in the test block is of a quality which will meet the abrasion test requirements.

4.6 **Rejected lots.**—Rejected lots may be offered again for inspection, provided the contractor has removed all nonconforming drills. The inspector shall again select and examine samples from such resubmitted lots to verify compliance with this specification.

5. PREPARATION FOR DELIVERY

(Civil agencies. — Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

5.1 Cleaning, preservation and packaging.

5.1.1 *Level A.*—Drills shall be cleaned, preserved and packaged in accordance with level A of Military Specification MIL–P–15424.

5.1.2 *Level C.*—Cleaning, preservation and packaging shall be in accordance with the contractor's commercial practice.

5.2 Packing.

5.2.1 *Level A.*—Drills cleaned, preserved and packaged level A or C as specified (see 6.1) shall be packed in accordance with the level A requirements of Military Specification MIL–P–15424.

5.2.2 *Level B.*—Drills cleaned, preserved and packaged level A or C as specified (see 6.1) shall be packed in accordance with the level B requirements of Military Specification MIL–P–15424.

5.2.3 *Level C.*—Drills cleaned, preserved and packaged as specified shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* — Marking for shipment shall be in accordance with the standard marking of Federal Standard No. 123.

5.3.2 *Military.*—In addition to any special marking required by the contract or order, or herein, interior and exterior shipping containers shall be marked in accordance with Military Specification MIL–P–15424.

6. NOTES

6.1 **Ordering data.**—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Size (nominal) and shank diameter (see table I).

(*c*) Single point carbide tipped pilot drill, when required (see 3.3).

(*d*) Selection of applicable levels of preservation, packaging and packing required (see 5.1 and 5.2).

(*e*) Marking, if different (see 5.3).

6.2 **Transportation description.**—Transportation description applicable to this item is:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

Tools, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.
Motor volume minimum weight 30,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:
Army—Ordnance Corps
Navy—Bureau of Ships
Air Force.

Copies of this specification may be purchased for 5 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.